Name:_____R-Number:_____

ECE4375/5375-001 Microprocessor Architecture
Project 2

NOTE: Submit the completed project folder as a zip file with your Verilog file and testbench by the due date specified on Canvas. Thoroughly comment your code and the testbench file. The uploaded zip file must be named as follows: firstLast-Project2.zip. The first comments to appear in all your code/testbench files must be the following:

```
/*

Name: First Last names
R-Number: R-XXXX…
Assignment: Project 2

*/
```

Replace the highlighted parts with the relevant information.

Write Verilog code to create a 32x32 register file. The register file should have two output busses (busA and busB), along with their corresponding bus addressing lines for each bus. The register file must allow loading the registers one at a time through an input data bus (Bus "D"), bus D address lines (DA), and register load signal (RL). Provide a working test bench as proof that your project is working. Test all registers for read (busA and busB) and write capability.